Session 2 - Method-Fit Matrix

Question: how should a predictive decision-support artifact and its economic usefulness be developed and evaluated in one sublevel-stoping operation?

The alternatives are research strategies, not algorithms. Scores and weights are proposed judgments for student review.

Criterion	Weight	A: design science with quantitative evaluation	B: longitudinal quantitative case study without a recommender	C: quasi-experimental operational change using current tools
E: epistemological fit	30	5	4	4
D: data availability	20	4	4	3
F: feasibility	25	3	4	2
C: contribution type	15	5	3	4
V: venue fit	10	4	4	4
Gate: E <= 2 eliminates	-	Survives	Survives	Survives
Weighted mean /5	100	4.20	3.85	3.30

Proposed calculation: sum(weight × score)/100, after applying the gate. This formula is explicit here and is not attributed as a dictated course formula.

A: E5/C5 reflect the artifact objective; D4 reflects reported data with quality/permissions unverified; F3 reflects development and authorization demands; V4 reflects thematic precedent, not editorial acceptance.

B: E4 supports a coherent association study; D4/F4 reflect lower intervention demands, still requiring academic-use permission; C3 would not deliver the entire intended artifact; V4 is preliminary thematic fit.

C: E4/C4 support studying an operational change; D3 depends on future interventions; F2 reflects authorization and confounding difficulties; V4 depends on methodological quality.

Proposed selection: A. B remains a legitimate lower-intervention alternative; C can become an evaluation phase. The matrix does not imply that ML must outperform a simpler model. Highest weight: epistemological fit, because the strategy must match the knowledge/artifact sought. Lowest weight: venue fit, because coherence and feasibility precede editorial preference.

Open tension: reassess feasibility if trials are denied. Equal-weight sensitivity yields A4.2, B3.8 and C3.4; this does not establish universal robustness.

Source: Session 2 PPT, internal slides 30, 39 and 40.